Cost Model: Replacement Only

Only the cost definition and its normalization implication are included.

Replace the Cost paragraph in Section 3.1.2.

Batch reservation cost. All M VMs are acquired at time zero and remain reserved until the batch completes at $H_{\max}(s)$. A VM that finishes its own queue early, or receives no task, is still charged until this common release. There is no separate VM release, provisioning delay, minimum charge, or billing-time rounding. Let r_j be the synthetic rate in cost units per second. The batch reservation cost is

$$\begin{aligned} C(s) &= H_{\max}(s) \sum_{j=1}^{M} r_j \\ &= \sum_{j=1}^{M} r_j H_j(s) + \sum_{j=1}^{M} r_j [H_{\max}(s) - H_j(s)]. \end{aligned} \tag{3}$$

The two terms on the second line charge busy time and reserved idle time. This is a fixed-fleet, continuous-time adaptation of reservation charging [1], rather than the full BaTS model or ERTH's resource-bandwidth cost [2].

For configured positive processing-rate bounds $\nu_{\min} \le \nu_j \le \nu_{\max}$, a denominator floor $\epsilon_\nu > 0$ in MIPS, and $0 \le r_{\min} \le r_{\max}$, the retained price mapping is

$$r_j = r_{\min} + (r_{\max} - r_{\min}) \sqrt{\frac{\nu_j - \nu_{\min}}{\max\{\nu_{\max} - \nu_{\min}, \epsilon_\nu\}}}. \tag{4}$$

The bounds belong to the configured capacity profile. The mapping is nondecreasing but not proportional to capacity, and it is synthetic rather than a calibrated cloud tariff. Cost depends on the batch horizon and fleet rates, not on per-task FCFS waiting. The earlier completion-time proxy $\sum_i r_{j(i)} CT_i$ is retained only as an exploratory control and is not used in the revised tables or figures.

Necessary normalization note (Section 3.1.3).

Under the adopted common-release policy, $R = \sum_j r_j$ is fixed and positive within an instance. When the reference floors are inactive, the normalized cost and makespan satisfy

$$\frac{C(s)}{C(s_g)} = \frac{R H_{\max}(s)}{R H_{\max}(s_g)} = \frac{H_{\max}(s)}{H_{\max}(s_g)}.$$

Consequently the four-component scalarization is equivalent to

$$F(s) = (w_{\text{ms}} + w_{\text{cost}}) \frac{H_{\max}(s)}{c_{\text{ms}}} + w_{\text{en}} \frac{E(s)}{c_{\text{en}}} + w_{\text{lb}} \frac{\mathcal{I}(s)}{c_{\text{lb}}}.$$

Nominal equal weights therefore give effective weights $(0.50, 0.25, 0.25)$ for makespan, energy, and imbalance. Cost remains a reported policy-dependent outcome but is not

an independent objective. Its local non-worsening test is redundant with the makespan test apart from numerical tolerances. Retaining the preceding nominal weights makes this change reproducible; it does not preserve the preceding effective preference.

References

[1] Oprescu, A.-M., Kielmann, T.: Bag-of-Tasks Scheduling under Budget Constraints. In: 2010 IEEE Second International Conference on Cloud Computing Technology and Science, pp. 351–359 (2010). https://www.cs.vu.nl/~amo/papers/bats.pdf.

[2] Qin X, Li S, Tong J, Xie C, Zhang X, Wu F, Xie Q, Ling Y, Lin G (2024) ERTH scheduler: enhanced red-tailed hawk algorithm for multi-cost optimization in cloud task scheduling. Artif Intell Rev 57:328. https://doi.org/10.1007/s10462-024-10945-6